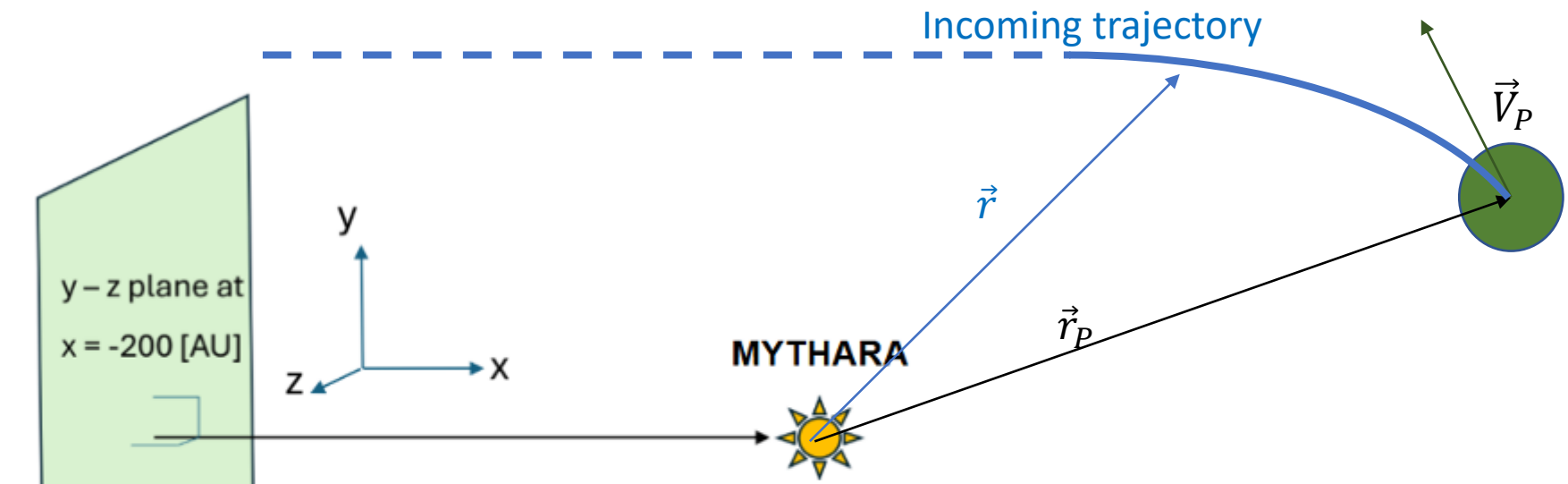




Parameter	Tolerance / Value
Conic Position & Velocity tolerance	100 m; 0.1 mm/s
Initial (trajectory start) Position & Velocity tolerance	100 m; 0.1 mm/s
V_{∞} equality tolerance	0.1 mm/s
Pericenter passage altitude	$0.1 R_{PL} \leq h \leq 100 R_{PL}$

h is altitude of pericenter passage, R_{PL} planetary radius

δ is turn angle of hyperbolic passage

For flybys («zero time flybys»)

Time Entering flyby and exiting flyby are the same

$$t_{IN} = t_{OUT}$$

Position Entering flyby

$$\vec{r}_{IN} = \vec{r}_P$$

Position exiting flyby

$$\vec{r}_{OUT} = \vec{r}_P$$

$$\vec{r}_{IN} = \vec{r}_{OUT} \quad |\vec{r}_{IN}| = |\vec{r}_{OUT}|$$

Inertial Velocity Entering flyby

$$\vec{V}_{IN} = \vec{V}_P + \vec{V}_{\infty IN}$$

Inertial Velocity Exiting flyby

$$\vec{V}_{OUT} = \vec{V}_P + \vec{V}_{\infty OUT}$$

$$|\vec{V}_{\infty IN}| = |\vec{V}_{\infty OUT}| = V_{\infty}$$

$$\vec{V}_{\infty IN} \cdot \vec{V}_{\infty OUT} = V_{\infty}^2 \cos \delta$$

$$\sin \frac{\delta}{2} = \frac{1}{e} = \frac{1}{1 + \frac{V_{\infty}^2 (R_{PL} + h)}{\mu_P}}$$